BIO-DATA

Name	**DR. SHWETA RAI**	
Designation	Assistant Professor- Department of Geology	
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Google Scholar ID	https://scholar.google.com/citations?user=_JIKmMIAAAAJ&hl=en	
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Educational Qualifications

Degree	Institution	Year
Ph.D. Geology	Department of Geology, Banaras Hindu University, India	2017-2024
M.Sc. (Tech.) Geology	Department of Geology, Banaras Hindu University, India	2014-2017

B.Sc. (Hons.) Geology	Faculty of Science, Banaras Hindu University, India	2011-2014

Permanent/ Temporary/ Contractual/ Adhoc Teaching Experience

Designation	Institution	Duration	From	To	Nature
Assistant Professor	Ram Lal Anand College, University of Delhi	___--	29.09.2025	Till Date	Permanent

Industry Experience (If any)

Designation	Company	Duration	From	To

Research Experience

Designation	Institution	From	To

Administrative Assignments

1. Member, College Event Management and Function Committee, Ram Lal Anand College, University of Delhi
2. Member, Eco Club (Vasudha), Ram Lal Anand College, University of Delhi

Areas of Interest / Specialization

Critical minerals and REEs; Clean Coal Energy; Coal Geology

Subjects Taught

Fuel Geology, Palaeontology, Research Methods in Geosciences, Techniques of Sample Collection & Processing in Geology, Crustal Evolution with Time, Greenbelt Development

Research Guidance (if any)

Research Projects (Major Grants/Research Collaboration)

Publications

1. Upadhayay, N. K., Naik, A. S., **Rai, S.,** Singh, P. K., Kumar, A., Rahim, A., Hakimi, M.H., Kumar, G. & Rajak, P. K. (2025). Geochemical and petrological characteristics coupled with stable isotope of the Permian Gondwana coals from Tatapani–Ramkola Coalfield, Son–Mahanadi Basin, India: Insights for paleodepositional and paleoclimate conditions. ***International Journal of Coal Geology*****,** 104882.
2. **Rai S.,** Singh A. L., & Singh P. K. (2024). Demineralization of Coal from Johilla Coalfield using Neutrophilic Native Bacteria: A Novel Technique to obtain REE from Coal. ***Journal of Geological Society of India*****,** 100(4), 486-494.
3. **Rai S.,** Singh A. L., Singh P. K., Kumar S., & Huisingh D. (2023). Optimization of extrinsic parameters for beneficiation of coal with the help of Cronobacter sp. **International Journal of Coal Preparation and Utilization,** 1-16.
4. Tripathi A., Arora A., **Rai S.,** Naik A.S., & Singh P. K. (2023). Study of surface microstructures and associated minerals in the coals of Singrauli coalfield, M.P., (India): Industrial implications. ***Journal of scientific research.*** 67(3). 1-10.
5. Naik A. S., Singh V., Mohanty D., Majhi R. K., Kumar G., Upadhyay N., Kumari M. & **Rai S**. (2023). Organic Petrological Facets and the Evolution of Paleomire, Matasukh Lignite Deposits, Rajasthan, India. ***Journal of the Geological Society of India,*** 99(10), 1372-1378.
6. **Rai S.,** Rai A., Kumar K., Naik A. S., Singh N., Srivastava D. K., & Singh P. K. (2022). Study of micro-structures and their relation with occurrence of mineral matter in Ramagundam Coals, Godavari basin, India: implications on coal and hydrocarbon industries. ***Journal of the Geological Society of India*****,** 98, 88-92.
7. Kumar H., **Rai S.,** Rai A., Faheem A., Naik A. S., Rajak P. K., Srivastava D.K., Maurya D.S., Chatterjee D. & Singh P. K. (2022). Characterization of composite coal seam III–IV of Ramagundam coalfield, Godavari Basin (India): an attempt to evaluate CBM prospects. ***Arabian Journal of Geosciences***, 15(16), 1400.
8. Rajak P. K., Singh V. K., Kumar A., Singh V., Rai A., **Rai S.,** Singh K.N., Sharma M., Naik A.S., Mathur N. & Singh P. K. (2021). Study of Hydrocarbon Source Potential of Kapurdi Lignites of Barmer Basin, Rajasthan, Western India. ***Journal of the Geological Society of India***, 97, 836-842.
9. Rajak P. K., Singh V. K., Singh A. L., Kumar N., Kumar O. P., Singh V., Kumar A., Rai A., **Rai S.,** Naik A.S. & Singh P. K. (2020). Study of minerals and selected environmentally sensitive elements in Kapurdi lignites of Barmer Basin, Rajasthan, Western India: Implications to environment. ***Geosciences Journal,*** 24, 441-458.
10. Medunić G., Singh P. K., Singh A. L., Rai A., **Rai S.,** Jaiswal M. K., Obrenović Z., Petković Z. & Janeš M. (2019). Use of bacteria and synthetic zeolites in remediation of soil and water polluted with superhigh-organic-sulfur Raša coal (Raša Bay, North Adriatic, Croatia). ***Water,*** 11(7), 1419.

Patents

Chapters in books and e-chapters (Published)

Conference Organisation/ Presentations

1. Poster presentation in the first 'National Symposium on Critical Mineral Exploration-NSCME 2026' at IIT Dhanbad on the topic 'Geomicrobiological controls on Coal demineralization and Rare Earth Element mobilization: Insight from functional gene analysis'.
2. Oral presentation in Goldschmidt Conference- 2023 (9-14 July, Lyon, France) on the topic 'Native neutrophilic coal bacteria- A potential tool for extraction of trace elements and Rare Earth Elements'.
3. Poster presentation entitled 'Use of Bacteria for removal of selective hazardous metals from sulfur-rich coals of Croatia' in "Recent Trends in Earth Science Research" organized by Department of Geology, BHU, India (2019).

Awards and Fellowships

1. H.S. Pareek Award (2024), The Geological Society of India
2. CSIR-UGC NET (2017)
3. GATE (2017)
4. Summer Research Fellowship jointly sponsored by I.A.Sc.(Bengaluru), INSA (New Delhi) and NASI (Allahabad) (2015)

Association With Professional Bodies

Other Activities

Dr. Shweta Rai

Assistant Professor

Department of Geology

Ram Lal Anand College